Battery Model Mathematical Formulation

Objective Function

$$\min \sum_{t \in T} \text{price}_t \; P_t \quad \text{..........} (1)$$

Power Gating Constraint

$$P_t = P_{\max} \;\; (\alpha_t \quad u_t \quad \beta_t + \gamma_t \quad u_t \quad \delta_t) \quad \text{..........} (2)$$

Where the smoothing terms are defined as:

$$\alpha_t = 0.5\,(1 + \tanh\,(k(u_t - 0.0025)))$$
$$\beta_t = 0.5\,(1 + \tanh\,(k(0.995 - \text{SOC}_t)))$$
$$\gamma_t = 0.5\,(1 - \tanh\,(k(u_t + 0.0025)))$$
$$\delta_t = 0.5\,(1 + \tanh\,(k(\text{SOC}_t - 0.005)))$$

Actual Power and SOC Dynamics

$$P_{\text{actual},t} = \eta \quad P_t \quad \text{..........} (3)$$

$$\text{SOC}_t = \text{SOC}_{t-1} + \Delta t \quad \frac{P_{\text{actual},t}}{E_{\max}} \quad \forall t \in T, t > 0$$

$$\text{SOC}_0 = \text{SOC}_{\text{initial}} + \Delta t \quad \frac{P_{\text{actual},0}}{E_{\max}}$$

Decision Variables & Bounds

Four variables are defined at each time step $t = 0, \dots, T-1$, each with explicit physical bounds:

1

Control Input u_t

Normalized charge/discharge command.

$$u_t \in [-1, 1]$$

2

State of Charge SOC_t

Fractional energy level of the battery.

$$SOC_t \in [0, 1]$$

3

Grid Power P_t

Scheduled power exchange with the grid.

$$P_t \in [-P_{\max}, P_{\max}]$$

4

Effective Power P_t^{act}

Battery power after efficiency losses.

$$P_t^{\mathrm{act}} \in [-\eta\, P_{\max}, \eta\, P_{\max}]$$

Sign convention: $P_t > 0$ → grid import (cost); $P_t < 0$ → grid export (revenue).

Objective: Minimize Electricity Cost (1)

Minimize total energy procurement cost over the horizon T:

$$\min_{u,\,\mathrm{SOC},\,P,\,P^{\mathrm{act}}} \sum_{t=0}^{T-1} \pi_t P_t$$

where π_t is the electricity spot price at step t. The optimizer exploits low-price periods to charge and high-price periods to discharge, reducing net grid cost.

Variable Legend

- $\boldsymbol{\pi_t}$ — electricity price at time t
- $\boldsymbol{P_t > 0}$ — import; pay $\pi_t.Pt$
- $\boldsymbol{P_t < 0}$ — export; earn $|\pi_t.Pt|$
- $\boldsymbol{T}$ — optimization horizon length

Constraints: Smooth power mapping (2)

Non-differentiable switching at SOC limits is approximated by smooth tanh-based gates, enabling gradient-based solvers.

$$\sigma_k(x) = \tfrac{1}{2}(1 + \tanh(kx))$$

The gated power map becomes:

$$P_t = P_{\max}\Big(\underbrace{\sigma_k(u_t - \varepsilon)\,\sigma_k(0.995 - \mathrm{SOC}_t)}_{\text{charging branch}}\ u_t + \underbrace{\sigma_k(-u_t - \varepsilon)\,\sigma_k(\mathrm{SOC}_t - 0.005)}_{\text{discharging branch}}\ u_t\Big)$$

$\sigma_k(u_t - \varepsilon)$

Activates charging branch when $u_t > 0$

$\sigma_k(-u_t - \varepsilon)$

Activates discharging branch when $u_t < 0$

$\sigma_k(0.995 - SOC_t)$

Suppresses charging near full ($SOC \approx 1$)

$\sigma_k(SOC_t - 0.005)$

Suppresses discharging near empty ($SOC \approx 0$)

Parameters: dead-band offset $\varepsilon = 0.0025$; steepness k controls sharpness of the gate transition. Larger k approaches a hard switch.

Constraints: Efficiency Model & SOC Dynamics (3)

Efficiency Map

A factor η is applied to both charge and discharge:

$$P_t^{\text{act}} = \eta\, P_t$$

Implementation uses η = *0.9.*

State-of-Charge Recursion

With Δt = 3600 *s* and capacity E_{max}:

$$\text{SOC}_0 = \text{SOC}_{\text{init}} + \frac{\Delta t}{E_{\max}} P_0^{\text{act}}$$

$$\text{SOC}_t = \text{SOC}_{t-1} + \frac{\Delta t}{E_{\max}} P_t^{\text{act}}, \quad t = 1, \ldots, T - 1$$

Each step integrates effective battery power into the energy state, enforcing causality across the horizon.

Analysis of RL Battery Optimization Box Plot (32 Runs)

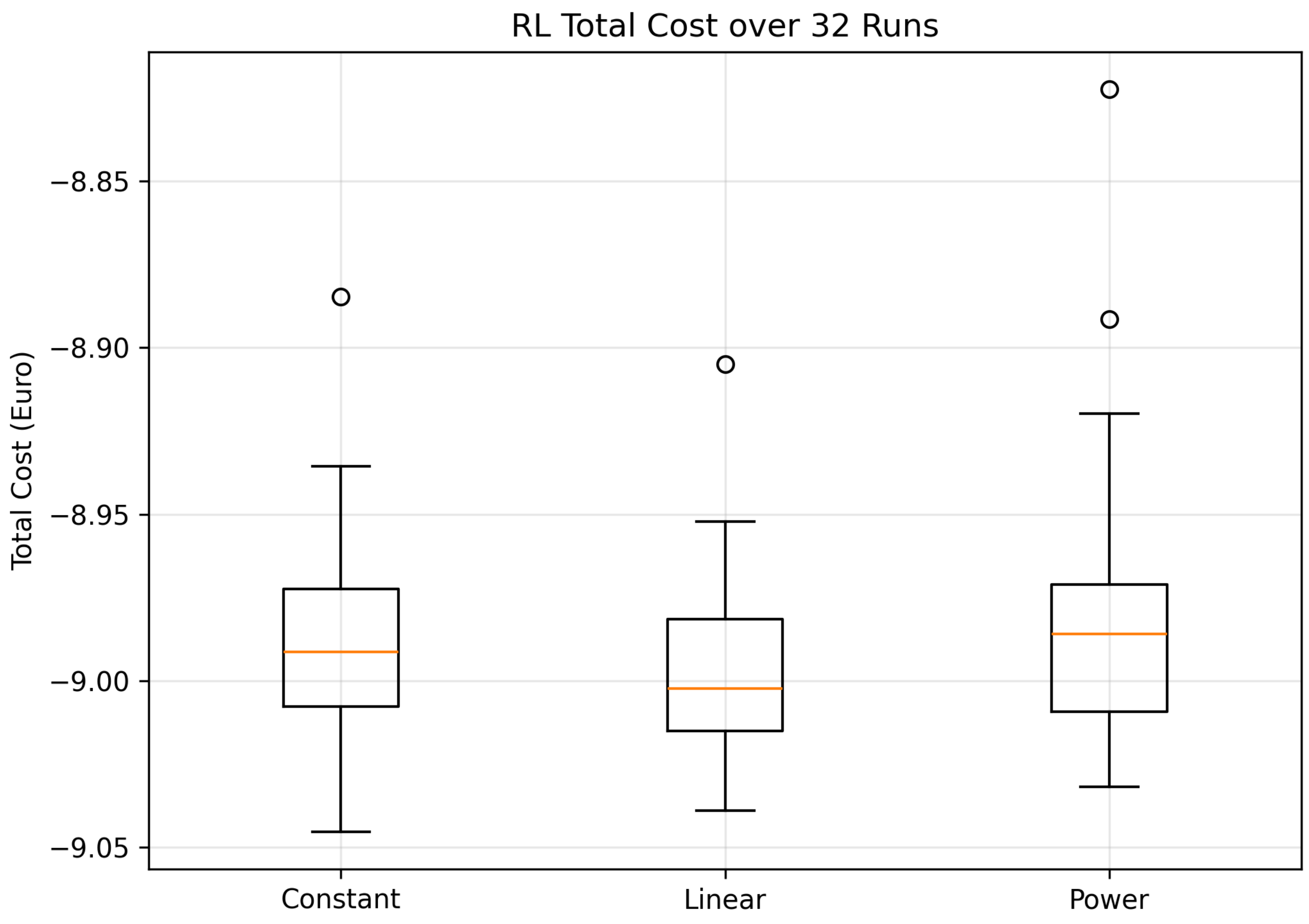


Fig.: Distribution of total cost from 32 RL models trained with identical hyperparameters and evaluated over the same time horizon.

* Mathematical Optimization Cost: -10.12 €

Analysis of RL Battery Optimization Box Plot (32 Runs)

The linear learning rate schedule performs best because it achieves the lowest median cost and maintains a tight distribution across runs, indicating stable and reliable convergence. The linear decay allows large updates during early exploration and smaller updates for fine policy refinement later in training.

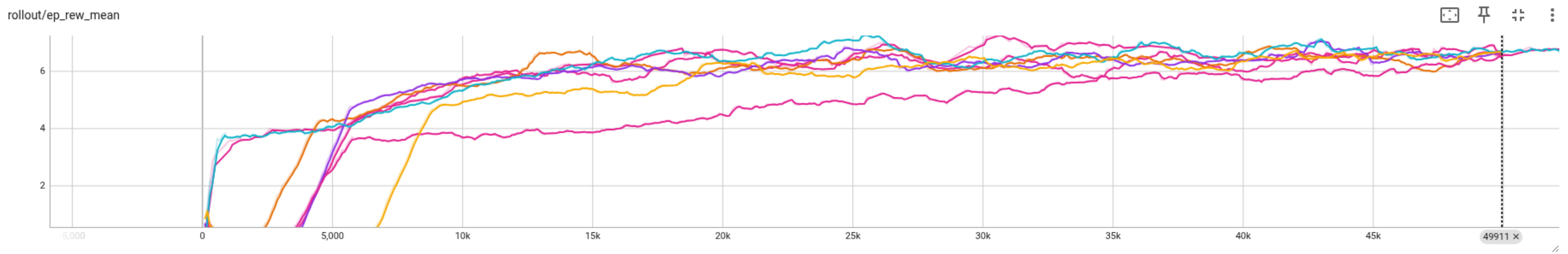
Strengths:

- 31/32 runs (96%) achieved near-optimal performance
- Stable and consistent learning behavior
- Model is robust to random initialization

Weaknesses:

- 4% chance of suboptimal convergence (1 outliers)
- Optimality of the learned policy cannot be guaranteed

Hyperparameter Tuning: Training Performance Comparison



Run	Smoothed ↓	Value	Step	Relative
SAC_2	6,7023	6,6986	49.920	18.08 min
optuna_trial_14/SAC_1	6,6698	6,6605	49.920	16 min
optuna_trial_6/SAC_1	6,6667	6,6524	49.920	18.91 min
optuna_trial_15/SAC_1	6,6517	6,6595	49.920	15.98 min
optuna_trial_17/SAC_1	6,5812	6,5399	49.920	7.023 min
SAC_3	6,5327	6,5	49.920	18.65 min
optuna_trial_13/SAC_1	6,5042	6,5004	49.920	15.89 min

Fig: Comparison of episode rewards across multiple SAC training runs with different hyperparameter configurations.

Computational Time Comparison

Method	Time
RL Training	≈ 16 minutes
RL Policy Execution	0.02 seconds
Optimization	0.16 seconds

Although RL requires higher upfront training time, the decision-making during operation is significantly faster than classical optimization, making RL suitable for real-time control applications.

Thank You!

I appreciate your time and attention.